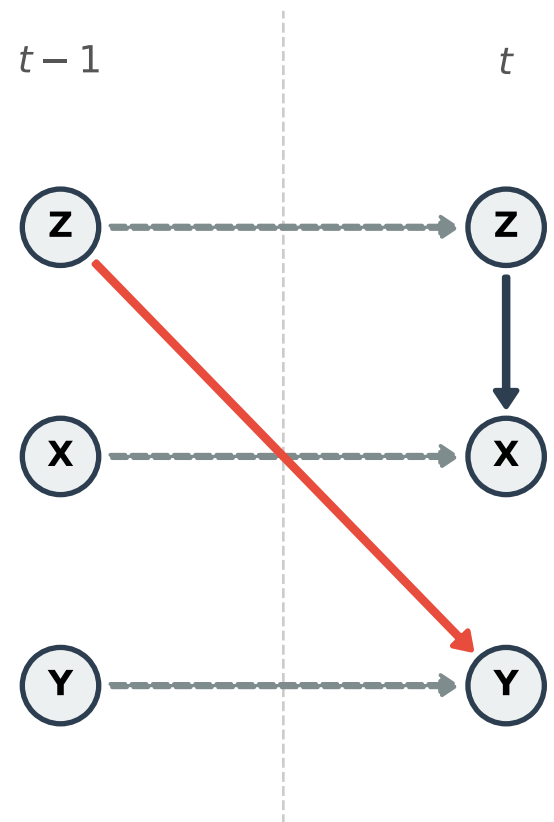


TSCM Sampler: Identifiability Case Study Structures

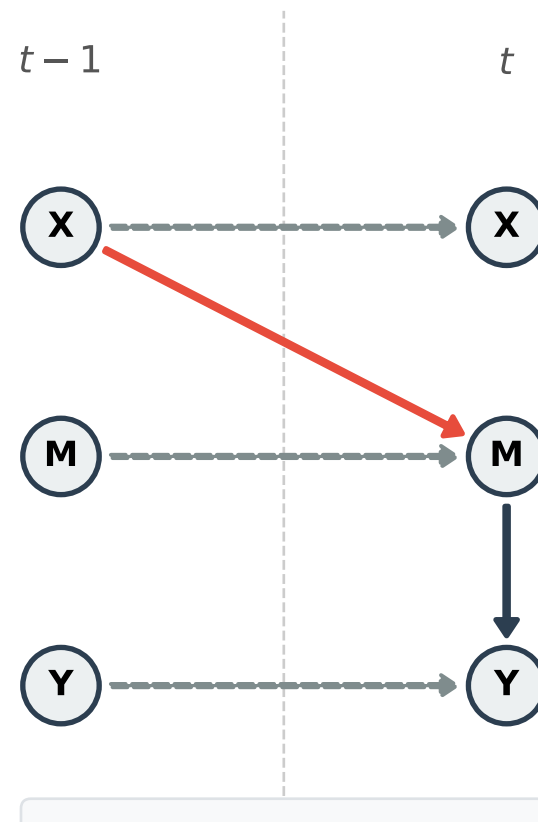
(a) Observed Confounder



$$p(y_t | do(x_t), H_{t-1}) = \int p(y_t | x_t, z_t) p(z_t | z_{t-1}) dz_t$$

Back-door adjustment through Z

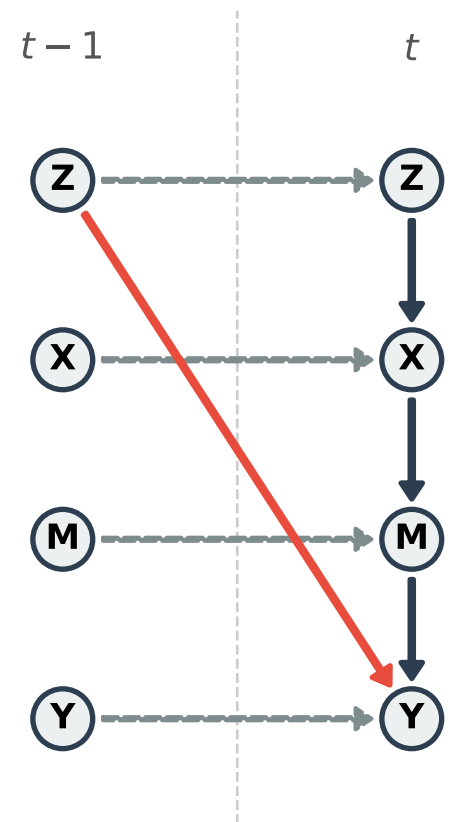
(b) Mediator



$$p(y_t | do(x_t)) = \sum_m p(y_t | m_t) p(m_t | x_{t-1})$$

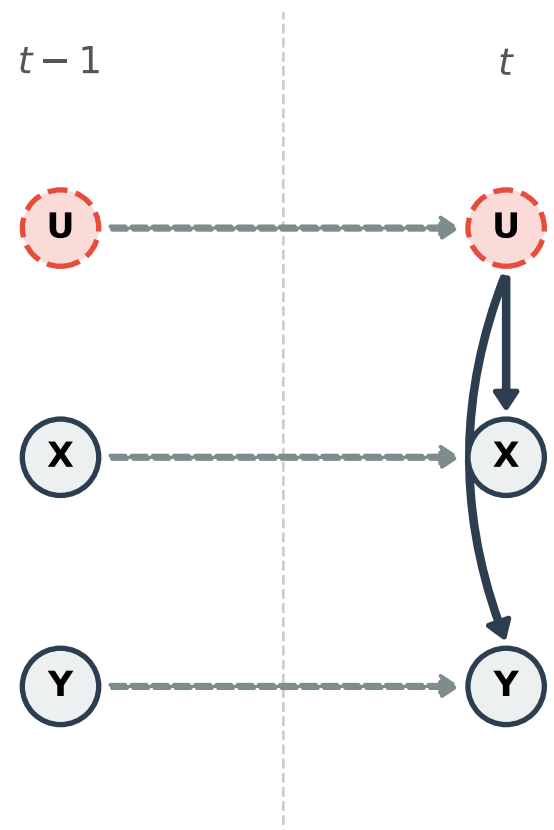
Causal effect fully mediated through M

(c) Confounder + Mediator



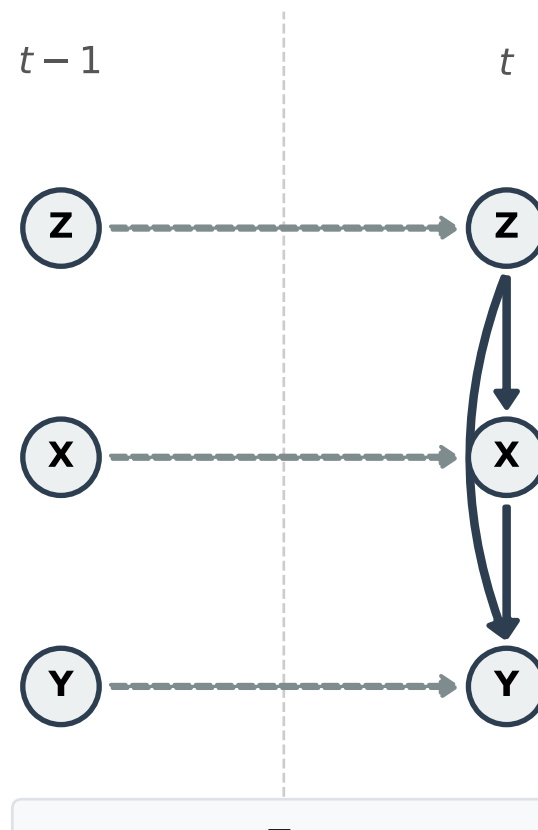
Hybrid: adjust for Z + trace through M
 $Z \rightarrow X \rightarrow M \rightarrow Y, Z(t-1) \rightarrow Y(t)$

(d) Unobserved Confounder



U confounds X and Y but is unobserved
 Not point-identifiable without instrument/mediator

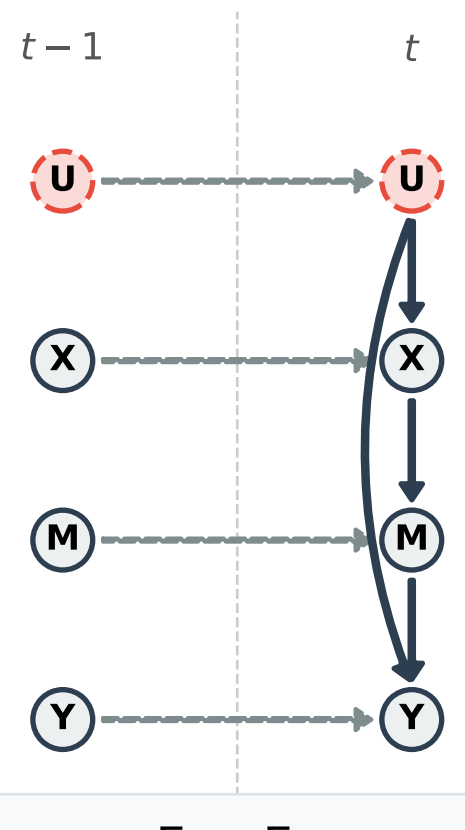
(e) Back-Door



$$p(y_t | do(x_t), H_{t-1}) = \sum_z p(y_t | x_t, z_t) p(z_t | H_{t-1})$$

Z satisfies the back-door criterion

(f) Front-Door



$$p(y | do(x)) = \sum_m p(m | x) \sum_{x'} p(y | x', m) p(x')$$

Front-door criterion through mediator M

Instantaneous edge Lagged edge Autoregressive Unobserved variable